

# Event Density Mobility Scaling in Concentrated BSA: FRAP Test of the Universal Degenerate-Mobility Law (Pre-Registered Protocol + Template Paper)

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## Event Density Mobility Scaling in Concentrated BSA: FRAP Test of the Universal Degenerate-Mobility Law

**Series:** Wave-3 Empirical-Prediction Sector — FRAP Results (TEMPLATE; placeholders pending experimental execution) **Status:** Template paper structured for completion after FRAP-BSA experimental execution (protocol active per `predictions/FRAP-High-BSA_InProcess/`; submitted to Creative Proteomics CIBR 2026-04-17). Data, results, and discussion sections contain placeholders marked [TO BE FILLED AFTER EXPERIMENT]. **This is not yet a results paper; it is the structure into which results will be filled when the experimental run completes.** **Date:** 2026-05-17 (template draft); experimental data acquisition pending **Authors:** Allen Proxmire (corresponding); [TO BE FILLED: experimental collaborator(s) if applicable] **Authors:** Universal Mobility Law Evidence paper (peer-reviewed, May 2026; 10-material UDM fit  $R^2 > 0.986$ ); FRAP-High-BSA\_InProcess protocol; Paper\_087 (13 primitives — Canon P4 Mobility Capacity Bound); Paper\_095 (form-FORCED / value-INHERITED methodology); Paper\_ED\_Predictions\_Bundle (companion bundle paper); Paper\_ED\_Postdictions\_PassedTests (UDM as passed test, §2).

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### §1 Introduction

The Universal Degenerate-Mobility (UDM) law derived from the Event Density (ED) substrate framework predicts a universal concentration-dependent diffusion functional form:

$$D(c) = D_0 \left(1 - \frac{c}{c_{\max}}\right)^\beta$$

with canonical  $\beta \approx 2$ . The functional form is **derived from substrate primitives** (ED Canon principle P4 — Mobility Capacity Bound) prior to any data fitting. The Universal Mobility Law Evidence paper (peer-reviewed, May 2026) confirmed this functional form across **10 materials** spanning soft-matter / biological systems with  $R^2 > 0.986$  at all materials. This established UDM as the corpus's first peer-reviewed external-validation-achieved result.

This paper reports the FRAP-BSA experimental test of UDM via an independent observable: **front-radius propagation exponent** in fluorescence recovery after photobleaching. The 2D porous-medium equation (PME) with  $m = \beta + 1 = 3$  derived from UDM with  $\beta = 2$  gives front-radius exponent:

$$\alpha_R = \frac{1}{d(m-1)+2} = \frac{1}{6} \approx 0.167.$$

ED predicts  $R(t) \sim t^{1/6}$  with a **sharp** bleach boundary (compact support, PME signature). The null prediction (standard Fickian diffusion) is  $R(t) \sim t^{1/2}$  with a Gaussian-blurred boundary.

The FRAP-BSA test is an **independent second confirmation** of UDM via a *different observable* (front propagation rather than diffusion-coefficient regression). A PASS here complements the existing peer-reviewed Universal Mobility Law evidence without replacing it; a FAIL would create tension with the published 10-material match and require resolution.

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## §2 Theory

### §2.1 The UDM mobility channel (ED Canon P4)

The ED PDE framework posits the substrate-level mobility-capacity bound (Canon P4): substrate-level mobility  $M(\rho)$  vanishes as the substrate event-density  $\rho$  approaches its maximum  $\rho_{\max}$ :

$$M(\rho) = M_0 (\rho_{\max} - \rho)^\beta.$$

Canonical  $\beta = 2$ . This is structural per ED Canon P4; the functional form is form-FORCED from substrate primitives prior to any empirical data.

### §2.2 Connection to concentration-dependent diffusion

The diffusion coefficient  $D(c) = M(c) \cdot \chi^{-1}(c)$ , where  $\chi(c) = dc/d\mu$  is the susceptibility. For concentration-dependent diffusion in the UDM regime,  $D(c) = D_0(1 - c/c_{\max})^\beta$  — the form confirmed in the peer-reviewed Universal Mobility Law paper across 10 materials.

### §2.3 Connection to the 2D porous-medium equation

For 2D radial spreading (FRAP geometry), the UDM mobility channel produces the 2D porous-medium equation (PME):

$$\frac{\partial c}{\partial t} = \nabla \cdot \left[ D_0(1 - c/c_{\max})^\beta \nabla c \right] = \nabla \cdot [D_{\text{eff}}(c) \nabla c]$$

with effective diffusion  $D_{\text{eff}}(c) \propto (1 - c/c_{\max})^\beta$ . This PME admits Barenblatt-type self-similar solutions with **compact support** (sharp front boundary) and front-radius scaling:

$$R(t) \sim t^{\alpha_R}, \quad \alpha_R = \frac{1}{d(m-1)+2}$$

with  $m = \beta + 1$  and  $d = 2$  (2D FRAP geometry). For  $\beta = 2$ :  $m = 3$ ,  $\alpha_R = 1/6 \approx 0.167$ .

## §2.4 ED prediction (form-FORCED) vs Fickian null (form-FORCED)

**ED-UDM prediction.** -  $R(t) \sim t^{1/6} \approx t^{0.167}$  - Sharp bleach boundary (PME compact support; characteristic of porous-medium-class diffusion) -  $R(t)$  slope on log-log plot:  $0.167 \pm 0.04$  (allowing for finite-sample fit uncertainty)

**Fickian null prediction.** -  $R(t) \sim t^{1/2} \approx t^{0.500}$  - Gaussian-blurred boundary (Fickian / heat-equation signature) -  $R(t)$  slope on log-log plot:  $0.500 \pm$  (fit uncertainty)

**Discrimination.** The two predictions differ in  $\alpha_R$  by a factor of 3 — well outside any reasonable observational uncertainty in a properly-designed FRAP experiment. PME vs Fickian fit can be discriminated by AIC (Akaike Information Criterion) at standard observational precision.

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## §3 Experimental protocol

### §3.1 Sample preparation

[Reproduced from FRAP-High-BSA\_InProcess protocol.md §4-§5, the operational protocol submitted to Creative Proteomics CIBR on 2026-04-17.]

**Bovine Serum Albumin–Fluorescein Isothiocyanate (BSA-FITC):** Sigma-Aldrich A9771 or equivalent. Reconstituted in pH 7.4 PBS at four concentrations:

- **200 mg/mL** ( $\phi = 0.15$ )
- **250 mg/mL** ( $\phi = 0.18$ )
- **300 mg/mL** ( $\phi = 0.22$ )
- **350 mg/mL** ( $\phi = 0.25$ )

Sample volumes 100  $\mu$ L per concentration. Loaded into glass-bottom 96-well plate (No. 1.5 coverslip, 0.17 mm thickness).

### §3.2 Photobleaching parameters

- **Bleach laser:** 488 nm, full power ( $\sim 10$  mW at sample), 100 ms exposure
- **Bleach geometry:** circular ROI,  $r_0 = 10$   $\mu$ m radius
- **Bleach intensity reduction:** target 50–70% of pre-bleach fluorescence (avoid full bleach to preserve front detectability)

### §3.3 Imaging parameters

- **Microscope:** confocal (Zeiss LSM 880 or equivalent), 40 $\times$  oil objective NA 1.3
- **Pixel size:** 0.2  $\mu$ m/pixel
- **Frame rate:** 1 frame/second for first 60 s post-bleach (front-propagation regime); 1 frame/5 s for 60–600 s (slow-recovery regime)
- **Detection:** GaAsP detector, 500–550 nm bandpass

### §3.4 Replication

- **5 bleaches per concentration  $\times$  4 concentrations = 20 total acquisitions**
- Bleach ROIs separated by  $\geq 50$   $\mu$ m to avoid cross-talk
- Single session per concentration where possible to minimize between-session drift

### §3.5 Data export

- Full TIFF stacks per acquisition (16-bit, no compression)

- Metadata: acquisition timestamp, pixel size, frame rate, laser power, ROI coordinates
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## §4 Expected results

### §4.1 ED-UDM prediction (PASS scenario)

If UDM holds for concentrated BSA:

| Concentration | Expected $\alpha_R$ | Expected front character |
|---------------|---------------------|--------------------------|
| 200 mg/mL     | $0.167 \pm 0.04$    | Sharp                    |
| 250 mg/mL     | $0.167 \pm 0.04$    | Sharp                    |
| 300 mg/mL     | $0.167 \pm 0.04$    | Sharp                    |
| 350 mg/mL     | $0.167 \pm 0.04$    | Sharp                    |

**PASS criterion.**  $\alpha_R \in [0.125, 0.170]$  at  $\geq 3/4$  concentrations + PME-front fit beats Fickian-front fit via AIC ( $\Delta\text{AIC} > 4$  in favor of PME).

### §4.2 Fickian null prediction (FAIL-of-ED scenario)

If standard Fickian diffusion holds:

| Concentration | Expected $\alpha_R$ | Expected front character |
|---------------|---------------------|--------------------------|
| 200 mg/mL     | $0.500 \pm 0.05$    | Gaussian-blurred         |
| 250 mg/mL     | $0.500 \pm 0.05$    | Gaussian-blurred         |
| 300 mg/mL     | $0.500 \pm 0.05$    | Gaussian-blurred         |
| 350 mg/mL     | $0.500 \pm 0.05$    | Gaussian-blurred         |

**FAIL-of-ED criterion.**  $\alpha_R \geq 0.35$  or Fickian-front fit wins via AIC at all 4 concentrations.

### §4.3 Mixed / intermediate outcomes

Five-outcome decision tree (per protocol §10):

- **PASS** — ED-UDM confirmed (criterion §4.1)
  - **NEAR-PASS** —  $\alpha_R$  within  $[0.125, 0.170]$  at  $\geq 2/4$  concentrations but not  $\geq 3/4$ ; or PME fit beats Fickian by  $\Delta\text{AIC} \in (2, 4]$  rather than  $> 4$
  - **FAIL** — Fickian wins per §4.2
  - **UNDECIDABLE** —  $\alpha_R \in [0.17, 0.35]$  across concentrations; AIC  $|\Delta\text{AIC}| < 2$
  - **SPLIT** — different concentrations give different outcomes (e.g., PASS at high concentration, Fickian at low concentration) — would suggest UDM is an asymptotic-high-concentration regime, requiring substrate-research-frontier follow-on
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## §5 Data analysis pipeline

### §5.1 Front-radius extraction

[Implementation: `analysis/scripts/udm_frap/frap_udm_pipeline.py` per protocol §9.4]

1. **Frame registration.** Cross-correlation alignment against pre-bleach frame to correct stage drift.

2. **Background subtraction.** Per-frame uniform background from off-bleach ROI.
3. **Front detection.** For each post-bleach frame:
  - Radial profile  $I(r, t)$  from ROI center
  - Front position  $R(t)$  = radius at which  $I(r, t) = 0.5 \cdot (I_{\text{pre-bleach}} - I_{\text{bleachcenter}}) + I_{\text{bleachcenter}}$  (50%-recovery isoline)
4. **Front-character quantification.** Sharp-boundary metric:  $|dI/dr|_{\text{max}}$  at front; Gaussian-fit residual character.

### §5.2 $\alpha_R$ regression

5. **Log-log fit.** Linear regression of  $\log R(t)$  vs  $\log t$  over the front-propagation regime ( $t \in [1, 30]$  s post-bleach). Extract  $\alpha_R$  + standard error.
6. **Per-concentration  $\alpha_R$  aggregation.** 5 bleaches per concentration  $\rightarrow$  5  $\alpha_R$  values per concentration  $\rightarrow$  mean  $\bar{\alpha}_R$  + standard deviation.

### §5.3 PME vs Fickian model fit

7. **PME fit.** Numerical solution of 2D PME  $\partial c/\partial t = \nabla \cdot [D_0(1 - c/c_{\text{max}})^\beta \nabla c]$  with  $\beta = 2$ . Two free parameters per acquisition:  $D_0$  and  $c_{\text{max}}$ . Least-squares fit to  $I(r, t)$  data.
8. **Fickian fit.** Numerical solution of 2D heat equation  $\partial c/\partial t = D\nabla^2 c$ . One free parameter per acquisition:  $D$ . Least-squares fit to  $I(r, t)$  data.
9. **AIC comparison.**  $\text{AIC}_{\text{PME}} = 2 \cdot 2 - 2 \ln L_{\text{PME}}$  vs  $\text{AIC}_{\text{Fickian}} = 2 \cdot 1 - 2 \ln L_{\text{Fickian}}$ .  $\Delta \text{AIC} = \text{AIC}_{\text{Fickian}} - \text{AIC}_{\text{PME}}$  favors PME if positive.

### §5.4 Per-concentration and ensemble decision

10. **Per-concentration outcome.** Apply 5-outcome decision tree (§4.3) using  $\bar{\alpha}_R + \Delta \text{AIC}$  at each concentration.
11. **Ensemble decision.** PASS if 3/4 concentrations PASS individually; otherwise per the 5-outcome tree.

## §6 Results

[TO BE FILLED AFTER EXPERIMENT]

### §6.1 Acquisition summary

[TO BE FILLED: Date of experimental session(s); total acquisitions completed; any acquisitions excluded due to technical failure and reason]

### §6.2 Per-concentration $\alpha_R$ measurements

[TO BE FILLED: Table of  $\bar{\alpha}_R$  + standard deviation per concentration]

| Concentration<br>(mg/mL) | $\phi$ | $\bar{\alpha}_R$ (measured) | Standard<br>deviation | $n$ acquisitions |
|--------------------------|--------|-----------------------------|-----------------------|------------------|
| 200                      | 0.15   | [TBD]                       | [TBD]                 | [TBD]            |
| 250                      | 0.18   | [TBD]                       | [TBD]                 | [TBD]            |
| 300                      | 0.22   | [TBD]                       | [TBD]                 | [TBD]            |
| 350                      | 0.25   | [TBD]                       | [TBD]                 | [TBD]            |

### §6.3 PME vs Fickian model comparison

[TO BE FILLED: Per-concentration  $\Delta\text{AIC}$  values; ensemble  $\Delta\text{AIC}$ ]

| Concentration (mg/mL) | $\Delta\text{AIC}$ (Fickian PME) | Preferred model |
|-----------------------|----------------------------------|-----------------|
| 200                   | [TBD]                            | [TBD]           |
| 250                   | [TBD]                            | [TBD]           |
| 300                   | [TBD]                            | [TBD]           |
| 350                   | [TBD]                            | [TBD]           |

### §6.4 Front character

[TO BE FILLED: Sharp-vs-blurred boundary classification per concentration; quantitative front-character metrics]

### §6.5 Ensemble outcome

[TO BE FILLED: 5-outcome classification (PASS / NEAR-PASS / FAIL / UNDECIDABLE / SPLIT)]

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## §7 Discussion

[TO BE FILLED AFTER EXPERIMENT]

### §7.1 Comparison with ED-UDM prediction

[TO BE FILLED: Match or mismatch with predicted  $\alpha_R = 1/6$ ; discussion of any per-concentration anomalies]

### §7.2 Comparison with peer-reviewed Universal Mobility Law evidence

[TO BE FILLED: Consistency or tension with the 10-material UDM fit ( $R^2 > 0.986$ , May 2026 publication); implications for the universality claim]

### §7.3 Implications for ED's substrate-level framework

[TO BE FILLED: If PASS, second-confirmation status + advancement of Canon P4 substrate-level framework; if FAIL, scope of refutation and which substrate-level claim(s) are challenged]

### §7.4 Comparison with alternative models

[TO BE FILLED: Comparison with alternative concentration-dependent-diffusion models (e.g., hard-sphere obstruction models, mode-coupling theory predictions, simple power-law fits without PME structure)]

### §7.5 Systematic uncertainties

[TO BE FILLED: Identified systematic uncertainties from experimental execution; impact on  $\alpha_R$  measurement; estimate of total uncertainty budget]

## §7.6 Future work

[TO BE FILLED: Extensions identified during analysis; follow-on experiments suggested; protocol refinements for V2]

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## §8 Conclusion

[TO BE FILLED AFTER EXPERIMENT: 2-3 paragraph summary of outcome + significance for ED's substrate-level framework + implications for UDM as the corpus's strongest empirically-validated structural prediction]

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## §9 Acknowledgments

[TO BE FILLED: Experimental collaborator acknowledgment (Creative Proteomics CIBR if executed via CP route, or alternative facility/group); funding source if applicable; any reviewers]

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## §10 Methods supplement

The full operational protocol is reproduced from `predictions/FRAP-High-BSA_InProcess/protocol.md` (the exact text submitted to Creative Proteomics CIBR on 2026-04-17). Sections §4-§8 of the protocol = experimental protocol used in this paper §3. Sections §9-§12 of the protocol = analysis pipeline used in this paper §5. Section §13 of the protocol = identified V2 protocol refinements (to be addressed in any follow-on experimental round).

Code repository for analysis pipeline: [TO BE FILLED: Zenodo DOI for analysis code release; or GitHub link for `analysis/scripts/udm_frap/frap_udm_pipeline.py`]

Raw data deposition: [TO BE FILLED: Zenodo DOI for TIFF-stack data deposition]

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## §11 References

### ED corpus papers:

- Paper\_087 — 13 primitives (position paper; Canon P4 Mobility Capacity Bound)
- Paper\_095 — Form-FORCED / value-INHERITED methodology
- Universal Mobility Law Evidence paper (peer-reviewed, May 2026) — 10-material UDM fit
- Paper\_ED\_Predictions\_Bundle (companion paper) — full ED predictions inventory
- Paper\_ED\_Postdictions\_PassedTests (companion paper) — UDM as passed test (§2)

### Protocol reference:

- `predictions/FRAP-High-BSA_InProcess/protocol.md` — full operational protocol

### External references (FRAP literature; UDM-application class materials):

- [TO BE FILLED: Standard FRAP-method references — Axelrod et al. 1976; Sprague et al. 2004; Mueller et al. 2008]
- [TO BE FILLED: BSA-FITC concentrated-solution literature; prior diffusion measurements at  $\phi > 0.10$ ]
- [TO BE FILLED: Universal Mobility Law Evidence paper full citation when DOI available]

- [TO BE FILLED: Porous-medium-equation theoretical references — Barenblatt 1996; Vázquez 2007]

**Cross-references:**

- For the UDM functional form derivation from ED substrate primitives: Universal Mobility Law Evidence paper §2-§3
- For the substrate-graph Canon P4 statement: Paper\_087 §P04 (Bandwidth) + Canon principle P4 derivation in `theory/Architectural_Canon.md`
- For the corpus-level context of this test within ED’s empirical-prediction sector: Paper\_ED\_Predictions\_Bundle §7

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**End Paper\_ED\_FRAP\_Template.** (*Template paper structured for completion after FRAP-BSA experimental execution. Data, results, and discussion sections contain placeholders marked [TO BE FILLED AFTER EXPERIMENT].*)